# Quantum-Enhanced Spatio-Temporal Point Process

for Dengue Fever Prediction in Southeast Asia

| 53K+ | 8 | 99.46% | R2=0.78 |
|---|---|---|---|
| STPP EVENTS | COUNTRIES | SPATIAL CORR. | FORECAST R2 |

Quantum Computing | Public Health | Hackathon 2026

Quantum Dengue Team | May 30, 2026 | NVIDIA RTX 3090 · 6.3 min GPU

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# The Dengue Crisis

## A Growing Threat in Southeast Asia

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## The Prediction Challenge

Why Classical Models Fail on Dengue Data layout: default background: '#0B1120'

# Research Questions & Hypothesis

CORE HYPOTHESIS

*"Quantum generative models can learn the complex probability distribution of dengue outbreak patterns more efficiently than classical methods — with **exponential advantage potential** as quantum hardware matures."*

## Research Questions

**RQ1**

Can quantum generative models learn spatial outbreak probability distributions with high fidelity?

**RQ2**

Does quantum-augmented data improve spatio-temporal forecasting beyond classical baselines?

**RQ3**

How does the quantum-inspired pipeline scale vs classical augmentation?

## Project Objectives

- Design QBM + QGAN for disease outbreak augmentation
- Generate synthetic patterns preserving spatial autocorrelation
- Validate via MMD loss (QBM) and spatial correlation (QGAN)
- Benchmark against Transformer-based classical forecaster

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# Hybrid Pipeline Architecture

## Quantum Generator + Classical Forecaster

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Quantum Born Machine

Learning Spatial Probability Distributions

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# Grid QGAN v3

## Full Spatial Grid Generation

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# Experimental Results

## QBM Convergence + Dataset Overview

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Forecasting Performance

Transformer vs Classical Baselines

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# Limitations & Roadmap

## Honest Assessment

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# Conclusion